Financial Transactions & Loans Dashboard
Microsoft Power BI Project Report

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Course: B.Com (FinTech & AI), 2nd Year

Tool Used: Microsoft Power BI Desktop

1. Introduction

For this project, I built a Financial Transactions & Loans Dashboard in Power BI using a banking dataset containing 11 tables. The dataset includes customer details, account balances, loans, transactions, branches and other related information.

The main purpose of the dashboard was to convert raw data into a simple and interactive visual report. It helps analyse customer information, account activity, loan status and transaction performance using KPI cards, charts, tables and slicers.

2. Objectives

1. To analyse customer, account, loan and transaction information.
2. To identify important KPIs such as Total Customers, Active Accounts, Active Loans and Total Transaction Value.
3. To compare accounts by account type and loans by status.
4. To analyse transaction trends over time.
5. To create an interactive dashboard using slicers.
6. To use drill-down, tooltips and conditional formatting for better analysis.

3. Dataset Description

The dataset contains 11 tables:

- Customers
- Customer Types
- Accounts
- Account Types
- Account Statuses
- Loans
- Loan Statuses
- Transactions
- Transaction Types
- Branches
- Addresses

These tables contain information about customers, account balances, loan amounts, interest rates and transaction details.

4. Data Preparation

Before creating the visuals, I cleaned the dataset using Power Query Editor.

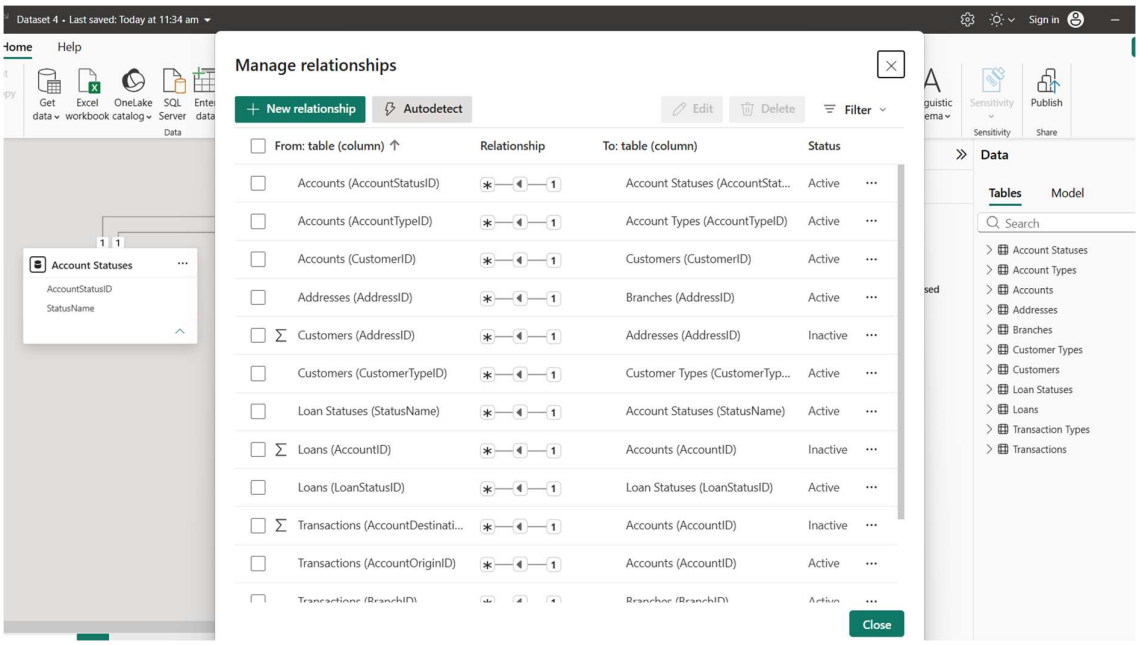
- Removed duplicate Account ID, CustomerID, Address ID and LoanID values.
- Changed date columns to the Date data type.
- Changed Balance, PrincipalAmount and Amount to Fixed Decimal Number.
- Replaced missing address and customer names with unknown.
- Corrected DOB formatting using the English (India) locale and replaced errors with null values.
- Applied Trim and Clean transformations to transaction descriptions.

These steps helped prepare the data for accurate analysis.

5. Relationships

I created 12 relationships between the tables using Model view. These relationships connect customers with accounts, accounts with loans, and transactions with transaction types and branches.

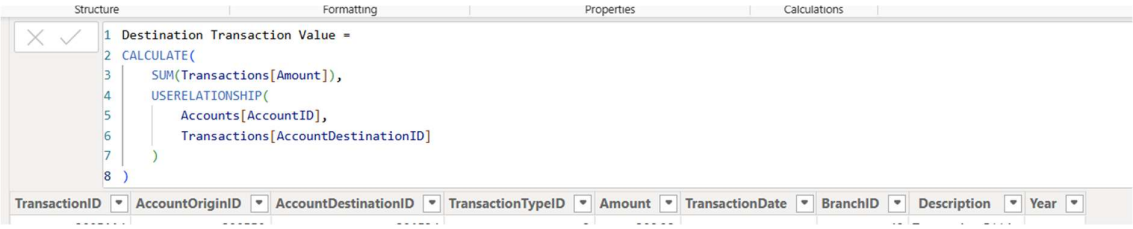
Some relationships were kept inactive, including the relationship between Accounts and Loans and the relationship between Accounts and Transactions through AccountDestinationID. These relationships can be activated within specific calculations using USERELATIONSHIP().



6. DAX Measures

I created the following measures to calculate the main financial indicators.

- Total Customers: Counts distinct customers.
- Total Accounts: Counts distinct accounts.
- Total Account Balance: Calculates the sum of account balances.
- Active Accounts: Counts accounts with Active status.
- Total Loans: Counts distinct loans.
- Total Loan Principal: Calculates the total loan principal.
- Average Interest Rate: Calculates the average interest rate.
- Active Loans: Counts loans with Active status.
- Total Transactions: Counts distinct transaction IDs.
- Total Transaction Value: Calculates the sum of transaction amounts.

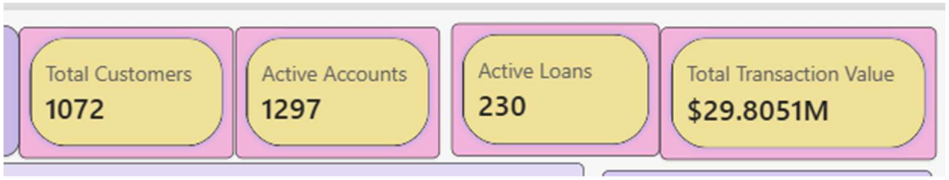


I also created Destination Transaction Value and Total Loan Amount measures using USERELATIONSHIP() to analyse data through inactive relationships.

7. Visuals Used

7.1 KPI Cards

I created four KPI cards displaying Total Customers, Active Accounts, Active Loans and Total Transaction Value. These cards provide a quick overview of the main financial indicators.



7.2 Accounts by Account Type

I used a Clustered Column Chart with Account Type on the X-axis and Total Accounts on the Y-axis. This helps compare the number of accounts across different categories.

7.3 Transactions by Month

I used a Line Chart with the Transaction Date hierarchy and Total Transactions. This helps analyse changes in transaction activity over time.

7.4 Loans by Loan Status

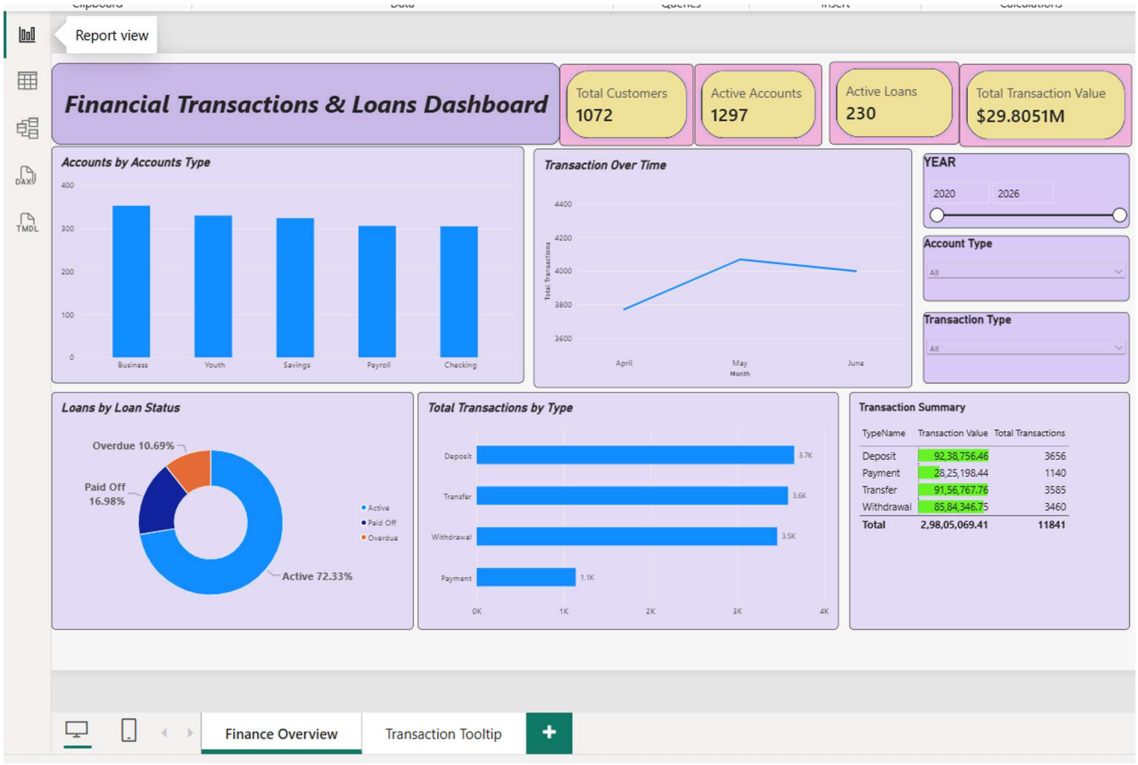
I used a Donut Chart with Loan Status as the category and Total Loans as the value. This helps compare the distribution of loans across different statuses.

7.5 Transactions by Type

I used a Clustered Bar Chart with Transaction Type and Total Transactions to compare transaction volumes across categories.

7.6 Transaction Summary

I created a Table visual containing Transaction Type, Total Transactions and Total Transaction Value. This provides a detailed comparison of transaction activity.



8. Slicers and Interactivity

I added three slicers to the dashboard:

- Year: Filters transactions by year.
- Account Type: Filters the report by account category.
- Transaction Type: Filters the report by transaction category.

I also used Format → Edit interactions to control how one visual affects another. This allows me to choose Filter, Highlight or None, depending on the required interaction.

9. Drill-Down

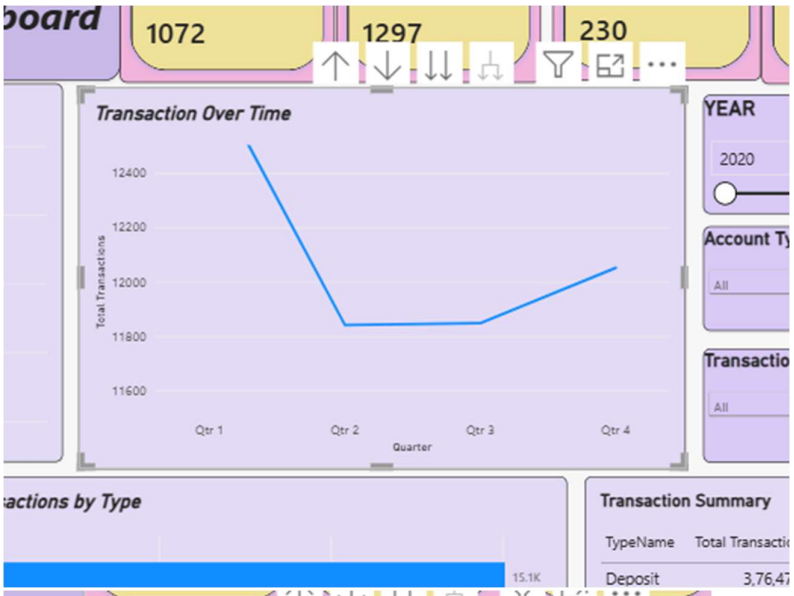
I created a line chart titled “Transactions Over Time” using the Transaction Date hierarchy on the X-axis and transaction details on the Y-axis. I used the drill-down feature to analyse transaction activity at three levels: year, quarter and month.

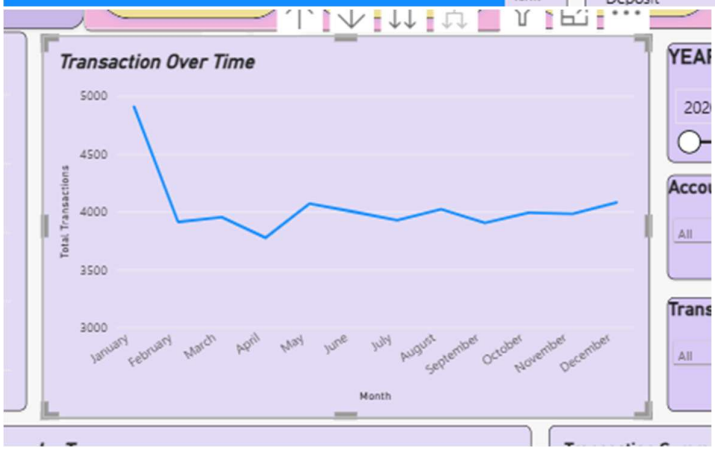
- Year Level: I set the X-axis to Year and adjusted the Y-axis range from 0 to 13,000 to view the overall transaction pattern.
- Quarter Level: I used the drill-down option to move from year to quarter and adjusted the Y-axis range from 11,500 to 12,500 to examine quarterly transaction activity.
- Month Level: I drilled down further to the month level and adjusted the Y-axis range from 3,000 to 5,000 to observe monthly transaction patterns.

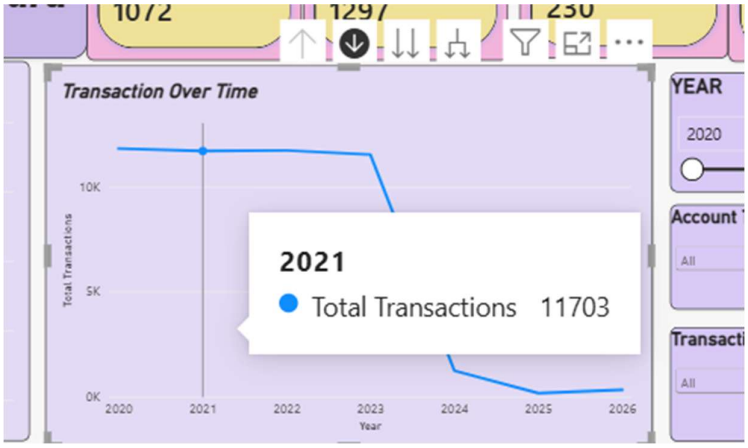
I also used the Drill Down mode to analyse a specific data point. I selected the year 2021, drilled down to Quarter 2, and then explored the months from April to June. This helped me examine transaction activity at different levels and understand the changes within a selected period.

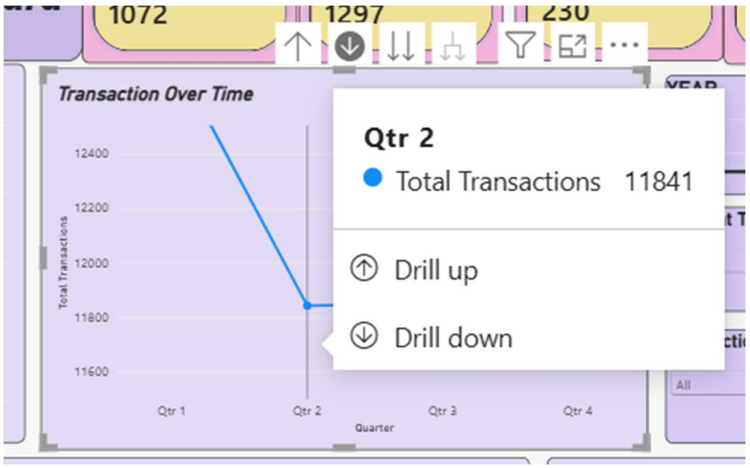
I captured screenshots of the Year, Quarter, Month and specific data-point drill-down views to document the analysis.

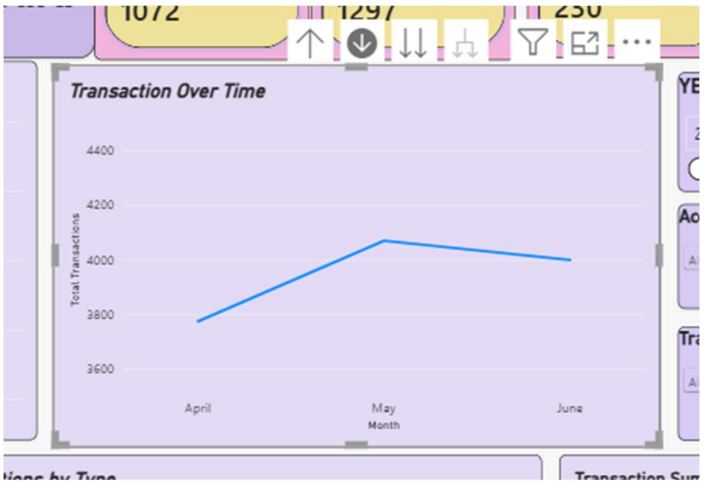



1072
1297
230
Transaction Over Time
12400
12200
12000
11800
11600
Total Transactions
Qtr 1
Qtr 2
Qtr 3
Qtr 4
Quarter
YEAR
2020
All
Transaction Summary
TypeName
Deposit


Transaction Over Time
5000
4500
4000
3500
3000
Total Transactions
January
February
March
April
May
June
July
August
September
October
November
December
Month
All


1072
1297
230
Transaction Over Time
10K
5K
0K
Total Transactions
2020
2021
2022
2023
2024
2025
2026
Year
2021
Total Transactions 11703
YEAR
2020
All

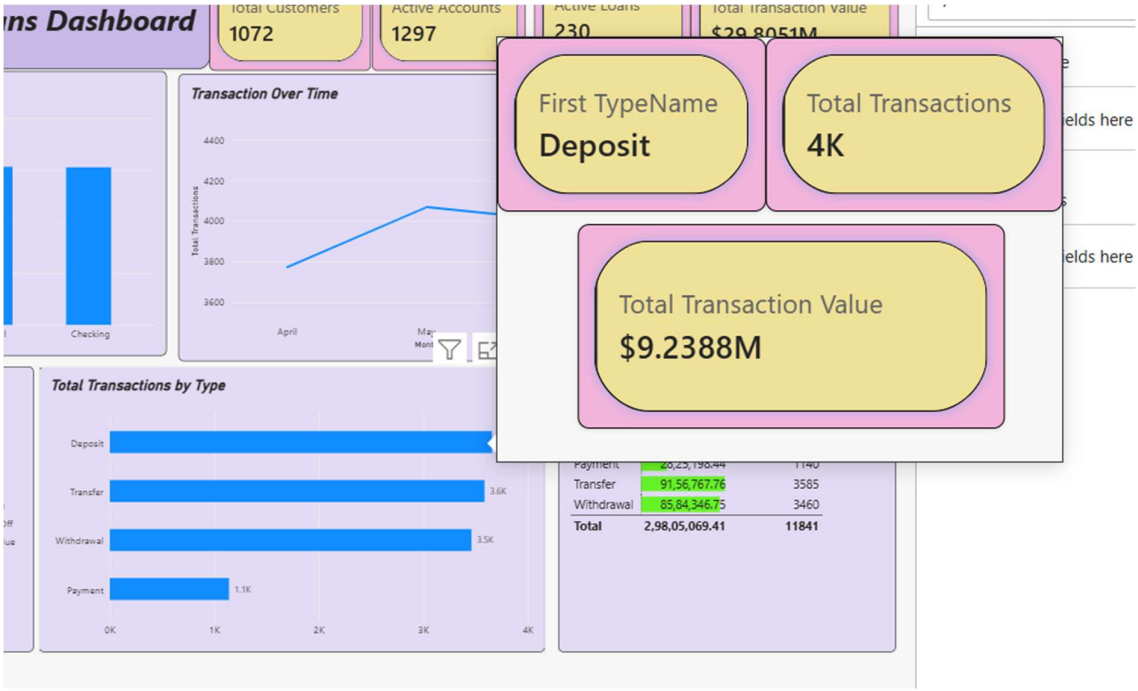

1072
1297
230
Transaction Over Time
Qtr 2
Total Transactions 11841
Drill up
Drill down
12400
12200
12000
11800
11600
Total Transactions
Qtr 1
Qtr 2
Qtr 3
Qtr 4
Quarter
All


1072
1297
230
Transaction Over Time
4400
4200
4000
3800
3600
Total Transactions
April
May
June
Month

10. Report-Page Tooltip

I created a separate page named Transaction Tooltip and enabled the tooltip option in Page information.

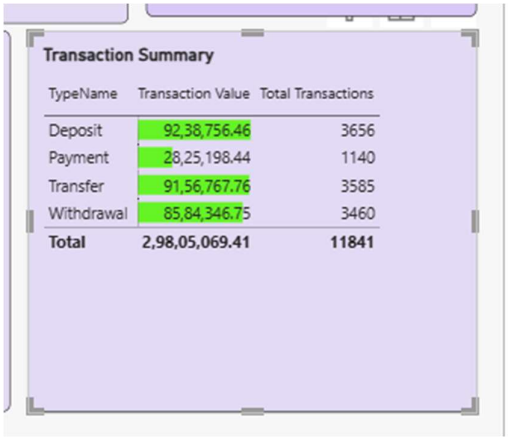
I added three cards displaying Transaction Type, Total Transactions and Total Transaction Value. I then connected this tooltip page to the Transactions by Type chart so users can view additional information when hovering over a data point.



11. Conditional Formatting

I applied data bars to the Total Transaction Value column in the Transaction Summary table. The formatting uses green for positive bars and red for negative bars, as specified in my project notes.

This helps users compare transaction values visually.



12. Key Findings

The dashboard is designed to help identify:

1. The total number of customers and accounts.
2. The number of active accounts and loans.
3. The account type with the highest number of accounts.
4. Changes in transaction counts over time.
5. The distribution of loans across statuses.
6. Differences in transaction counts and values across transaction types.

The actual numerical findings must be recorded after validating the results in Power BI.

Conclusion

Through this project, I learned how Power BI can be used to clean data, create relationships, develop DAX measures and build an interactive financial dashboard. The charts, KPI cards and slicers help analyse customers, accounts, loans and transactions from different perspectives.

Overall, this project gave me practical experience in data modelling, financial data analysis and dashboard development using Microsoft Power BI.